



Data Science & Artificial Intelligence



Machine Learning

Multi-layer Perceptron

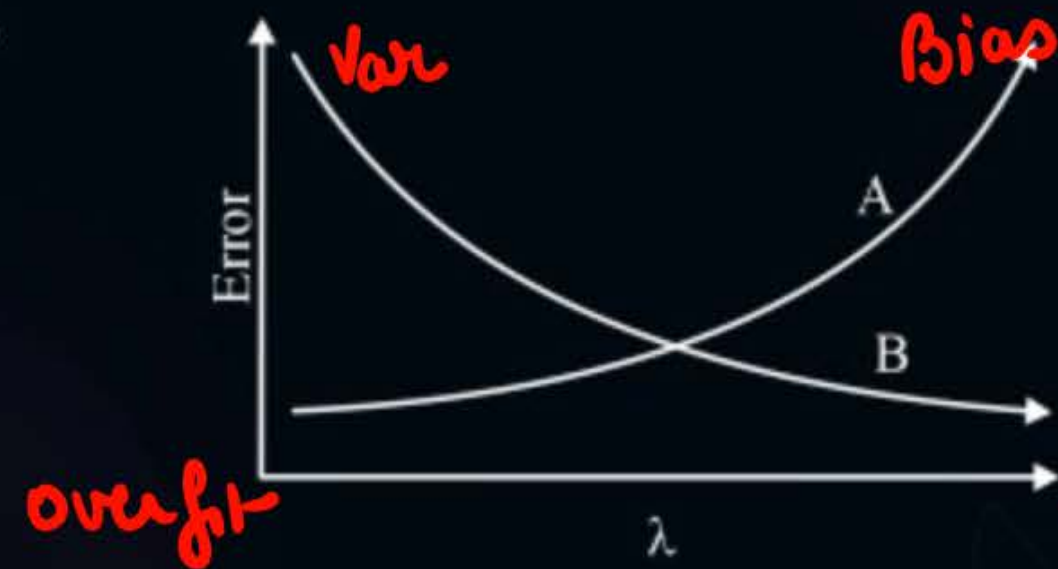


DPP 01 Discussion Notes

By – Siddharth Sabharwal Sir

[MCQ]

#Q. Assuming a regularization penalty of the form $\lambda R(\theta)$. Complete the following illustration. Note that the x-axis is the regularization parameter λ and not the model complexity.



• λ inc: Regularisation inc

• underfit

λ underfit

A

Is the test Error and (B) is the error due to $(\text{Bias})^2$.

B

Is the error due to Model variance and (B) is the Training Error.

C

Is the error due to Model variance and (B) is the error due to $(\text{Bias})^2$.

D

Is the error due to $(\text{Bias})^2$ and (B) is the error to model Variance.

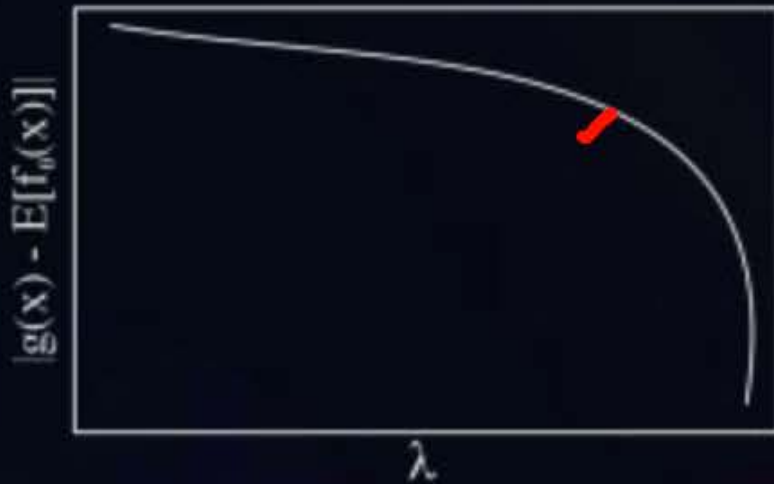
[MCQ]

#Q. Which of the following could plausibly be a plot of the magnitude (i.e., absolute value) of model bias as a function of λ if we use multiple linear regression with exponential regularization? ✓

Assume that values on the x-axis are linear from 0 asymptotically large λ .

Circle the letter corresponding to the most plausible plot.

A



λ inc
Bias dec

B



Not in Syllabus

C



D



[MSQ]

#Q. Ishani trains two model on the same data with the same features: Model A and Model B. Information on both models is given below:

Model	Bias Squared	Model Variance
A	2 ✓	12 ✓ ← overfit
B	4	1 ← underfit

Which of the following statements are true? Select at that apply.

- ☒ **A** Model A will overfit more than Model B for this dataset
 - Model B Reg, high λ
 - Model A OLS overfit.
- ☐ **B** If both models used LASSO Regression, Model B had the higher λ .
- ☐ **C** Model A could have used Ridge Regression with $\lambda > 0$ while Model B used OLS.
 - OLS
- ☐ **D** Model B must be larger weights than Model A.

[MCQ]

#Q. Matthew decides to try a new method. He first comes up with a model $h(x)$, which has a bias of B and a variance of V on the original dataset. Next, he builds a new model $f(x)$ using this process:

1. Obtain n independent, random samples from the original dataset.
2. For each dataset, train a new version of $h_i(x)$ for each i from 1 to n , where $h_1(x)$ was the $h(x)$ model trained on the first dataset and $h_n(x)$ was the $h(x)$ model trained on the n^{th} dataset.
3. Finally, define $f(x)$ as $f(x) = \sum_{i=1}^n h_i(x)$.

data $\rightarrow$ (break into n sets)
 $\rightarrow$ Train $h(x)$ from each set.

What is the model's bias squared of the $f(x)$ model in terms of B and n ? Grading will be done based on the work you shown in the box below.

A B^2 ✓

B $\frac{1}{n} B^2$

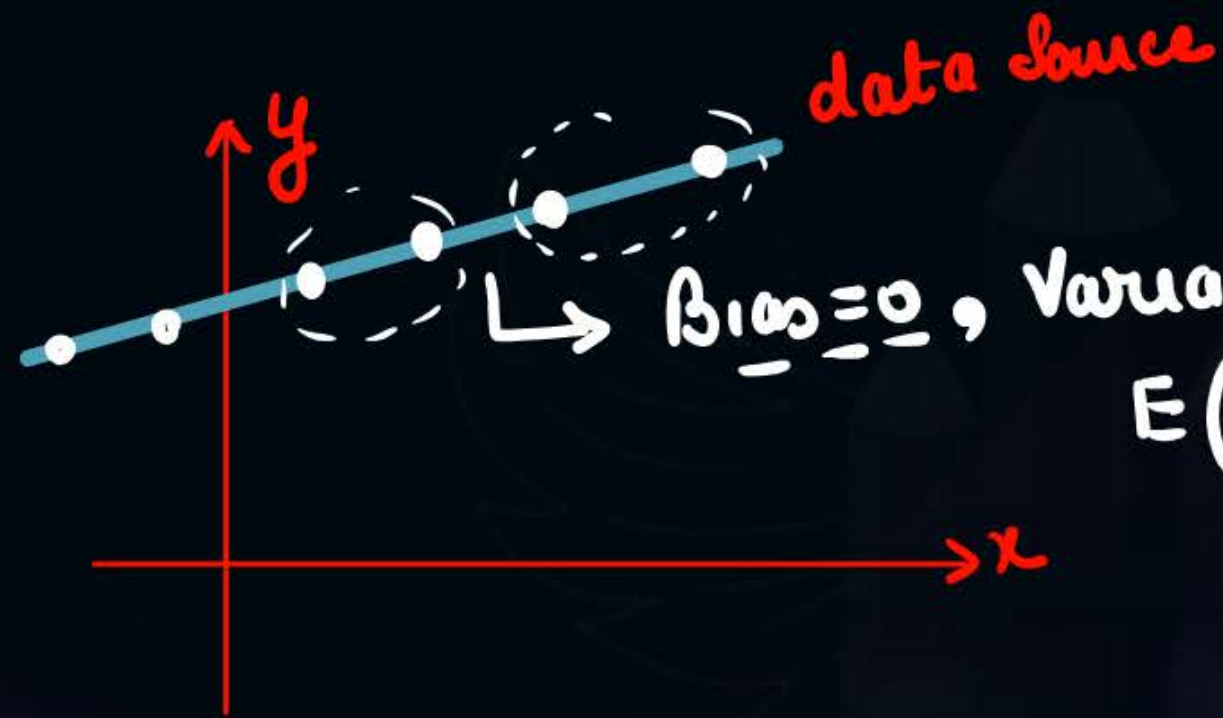
C $\left(\frac{B}{n}\right)^2$

D nB^2

[MCQ]

- #Q. If we assume that the data is being generated from a source with no noise
 ➔ $y = w^T x$, choosing a classifier $f^*(x) = w^T x$ using OLS will make which of the following quantities zero?

• data $\Rightarrow$ no noise



- A** Only Bias
- B** Only Variance
- C** ✓ Both Bias and Variance
- D** None of the above

$$\text{Bias} = 0, \text{ Variance} = 0$$

$$E \left(\hat{f}(x) - E(\hat{f}(x)) \right)^2$$

[NAT]

- #Q. Consider a Multi-Layer Perceptron (MLP) model with one hidden layer and one output layer. The hidden layer has 10 neurons, and the output layer has 3 neurons. The input to the MLP is a 5-dimensional vector. Each neuron is connected to every neuron in the previous layer, and a bias term is included for each neuron. The activation function used is the sigmoid function. Calculate the total number of trainable parameters in this MLP model.

$$\begin{array}{ccc}
 & 50 & \xrightarrow{10} 30 & \xrightarrow{3} \\
 5 & - & 10 & - & 3 \\
 \text{I/p} & & \text{H/L} & & \text{O/P}
 \end{array}
 \Rightarrow (93)$$

[NAT]

#Q. Consider the following binary classification dataset:

(x_1, x_2, x_3, y)

$(0, 0, 1, +1)$

$(0, 1, 1, -1)$

$(1, 0, 1, -1)$

x_3 is used for the bias term.

Learning Rate = 1

	x_1	x_2	Padding	
1	0	0	1	+1
2	0	1	1	-1
3	1	0	1	-1

Suppose we initialize the weight vector for a linear classifier to $w = (0, 0, 0)$.

This is the case of SLP. What are the weights after one pass through the data in the order listed above?

$$w = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix}$$

• We start with $w = 0, 0, 0$

$$1) y_i(w^T x_i + b) = 1(0 + 0 + 0) = 0 \Rightarrow \text{error}$$

$$w^{\text{new}} = w^{\text{old}} + \eta y_i x_i \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$2) y_i(w^T x_i + b) = -1(0 + 1) = -1 \Rightarrow \text{error}$$

$$w^{new} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 1 \cdot (-1) \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$w^{new} \Rightarrow \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$$3. y_i(w x_i + b) = -1(0 + 0 + 0) = 0 \text{ error}$$

$$w^{new} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} + (-1) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\underline{w^{new}} \Rightarrow \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$$

[MCQ]

#Q. Given the following training points for classification, where each point is represented as (x_1, x_2, y)

$(1, 0, +)$

$(1, 1, +)$

$(0, 1, -)$

no bias

$$1) y_i(w x_i) = 1 \times 0 = 0 \text{ error}$$

$$w^{new} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 1 \cdot (1) \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow w^{new} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Perform one pass of the perceptron algorithm on this data, starting from a weight vector initialized at $w = [0, 0]$ and using the decision rule $w^T x > 0$, (i.e., classifying as positive if the source is greater than 0.)

Assume on bias term is used.

What is the final weight vector w after this pass?

A $[1, -1]$

C $[1, 0]$

B $[2, 1]$

D $[0, 1]$

$$2. y_i(w x_i) = +1(1) = +1 \text{ no error}$$

$$3. y_i(w x_i) = -1(0) = 0 \text{ error}$$

$$\omega^{\text{new}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + (-1) \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\omega^{\text{new}} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

[NAT]

#Q. Assume that you are given observations $(x, y) \in \mathbb{R}^2 \times \{-1, +1\}$ in the following order:

Instance	1	2	3	4	5	6	7	8
Label y	+1	-1	+1	-1	+1	-1	+1 ✓	+1 ✓
Data (x_1, x_2)	(10, 10)	(0, 0)	(8, 4)	(3, 3)	(4, 8)	(0.5, 0.5)	(4, 3)	(2, 5)

Show the action of the perceptron algorithm for the above sequence of observations. We start with an initial set of weights $w = (1, 1)$ and bias $b = 0$.

$$1) y_i(w x_i + b) = 1(20 + 0) = +20 \Rightarrow \text{no error}$$

$$2) y_i(w x_i + b) = (-1)(0 + 0) = 0 = \text{error} \quad w^{\text{new}} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$3) y_i(w x_i + b) = 1(12 - 1) = +11 \Rightarrow \text{no error}$$

$$4) y_i(w x_i + b) = (-1)(5) = -5 \Rightarrow \text{error} \Rightarrow w^{\text{new}} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + (-1) \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix} \Rightarrow \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix}$$

$$5. y_i(\omega x_i + b) < 0 \text{ error}$$

$$\omega^{\text{new}} = \begin{pmatrix} -2 \\ -2 \\ -2 \end{pmatrix} + (1) \begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}$$

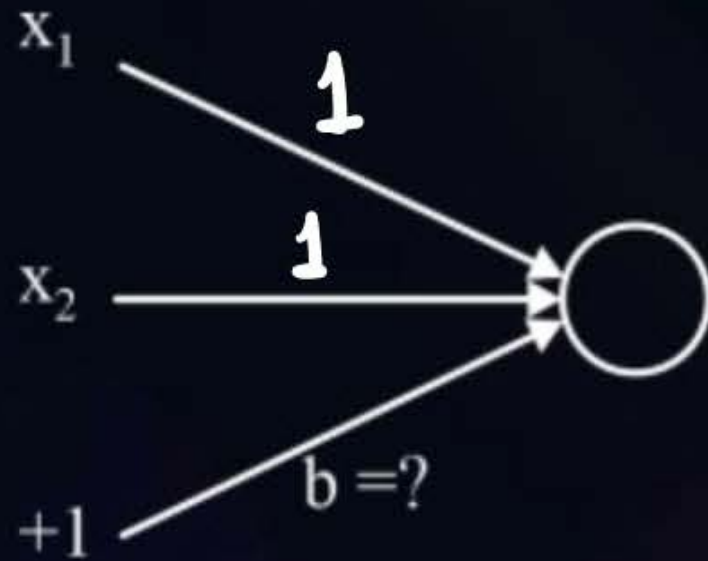
$$6. y_i(\omega x_i + b) = (-1)(1+3-1) < 0 \Rightarrow \text{error}$$

$$\omega^{\text{new}} = \begin{pmatrix} -1 \\ 2 \\ 6 \end{pmatrix} + (-1) \begin{pmatrix} 1 \\ .5 \\ .5 \end{pmatrix} = \begin{pmatrix} -2 \\ 1.5 \\ 5.5 \end{pmatrix}$$

$$\omega^{\text{new}} \Rightarrow \begin{bmatrix} -2 \\ 1.5 \\ 5.5 \end{bmatrix}$$

[NAT]

- #Q. Suppose we want to implement logical OR using perceptron.
For the given decision rule, what will be the bias term if weights are given as $w_1 = w_2 = 1$.
There are multiple possibilities, gives anyone.



OR		
X1	X2	Y
0	0	0
0	1	1
1	0	1
1	1	1

$$y = \begin{cases} 0, & \text{if } w \cdot x + b \leq 0 \\ 1, & \text{if } w \cdot x + b > 0 \end{cases}$$

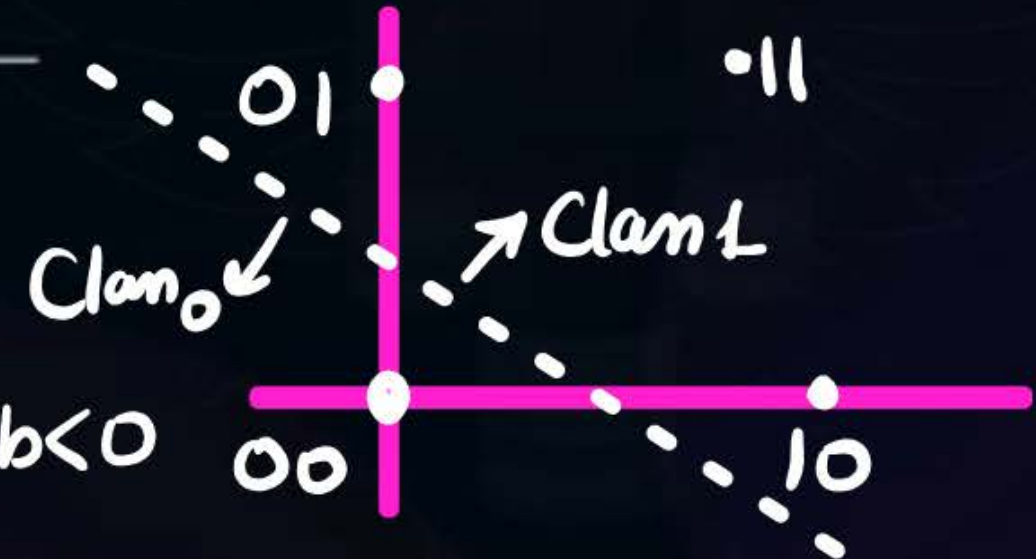
$$x_1 + x_2 + b = 0$$

b can be any value from $-1 < b < 0$

$$\text{~~~~~}$$

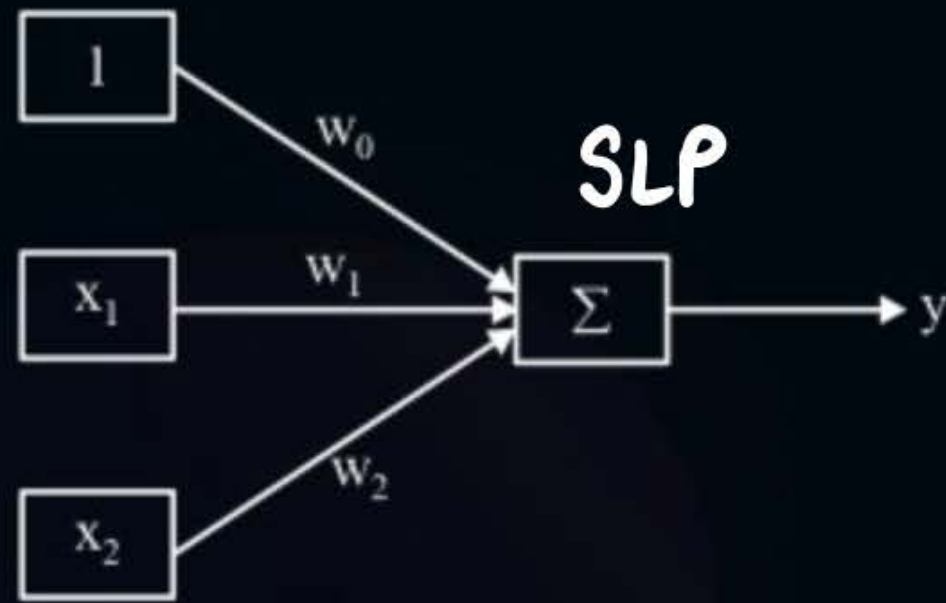
$$-1 < b < 0$$

$$\text{~~~~~}$$



[MCQ]

#Q. Consider the following perceptron, for which the inputs are the always 1 features and two binary features $x_1 \in \{0,1\}$ and $x_2 \in \{0,1\}$. The output $y \in \{0,1\}$.



$$y = \begin{cases} 1, & \text{if } (w_1 \cdot x_1 + w_2 \cdot x_2) > 0 \\ 0, & \text{otherwise} \end{cases}$$

↗ linear cl.

Which one(s) of the following choices for the weight vector $[w_0 \ w_1 \ w_2]$ can classify y as $y = (x_1 \text{ XOR } x_2)$? XOR is the logical exclusive or operation, which equals to zero when x_1 equals to x_2 and to one when x_1 is different from x_2 .

- A** [1 1 0]
- B** [1 1 0]
- C** [-2 1 1.5]
- D** Any weight that satisfy $(-w_1 - w_2) < w_0 < \min(0, -w_1, -w_2)$
- E** No weight can compute the XOR logical relation.



[NAT]

#Q. Consider a Perceptron that needs to classify binary input data correctly. The dataset includes the following input-output pairs, where each input has three binary values, and the output is a binary classification label:

(111, 1) (011, 1)

(110, 1) (010, 0)

(000, 0)

The rule for classification is: the output should be 1 if at least two of the three input are 1; otherwise, the output should be 0.

Can you determine a suitable set of weights and a threshold value for the perceptron so that it correctly classifies this dataset?

$$\checkmark x_1 + x_2 + x_3 \geq 2 \Rightarrow \text{Class 1}$$

$$\checkmark x_1 + x_2 + x_3 < 2 \Rightarrow \text{Class 0}$$

} Three input x_1, x_2, x_3
 $\bullet x_1 + x_2 + x_3 \geq 2 \bullet$

[MSQ]

#Q. Which of the following is/are true about the perceptron classification?

A, C

A

It can learn a OR function ✓

B

It can learn a XOR function ✗

$$w^{new} = w^{old} + \eta y_i x_i$$

C

The obtained separating hyperplane depends on the order in which the points are presented in the training process.

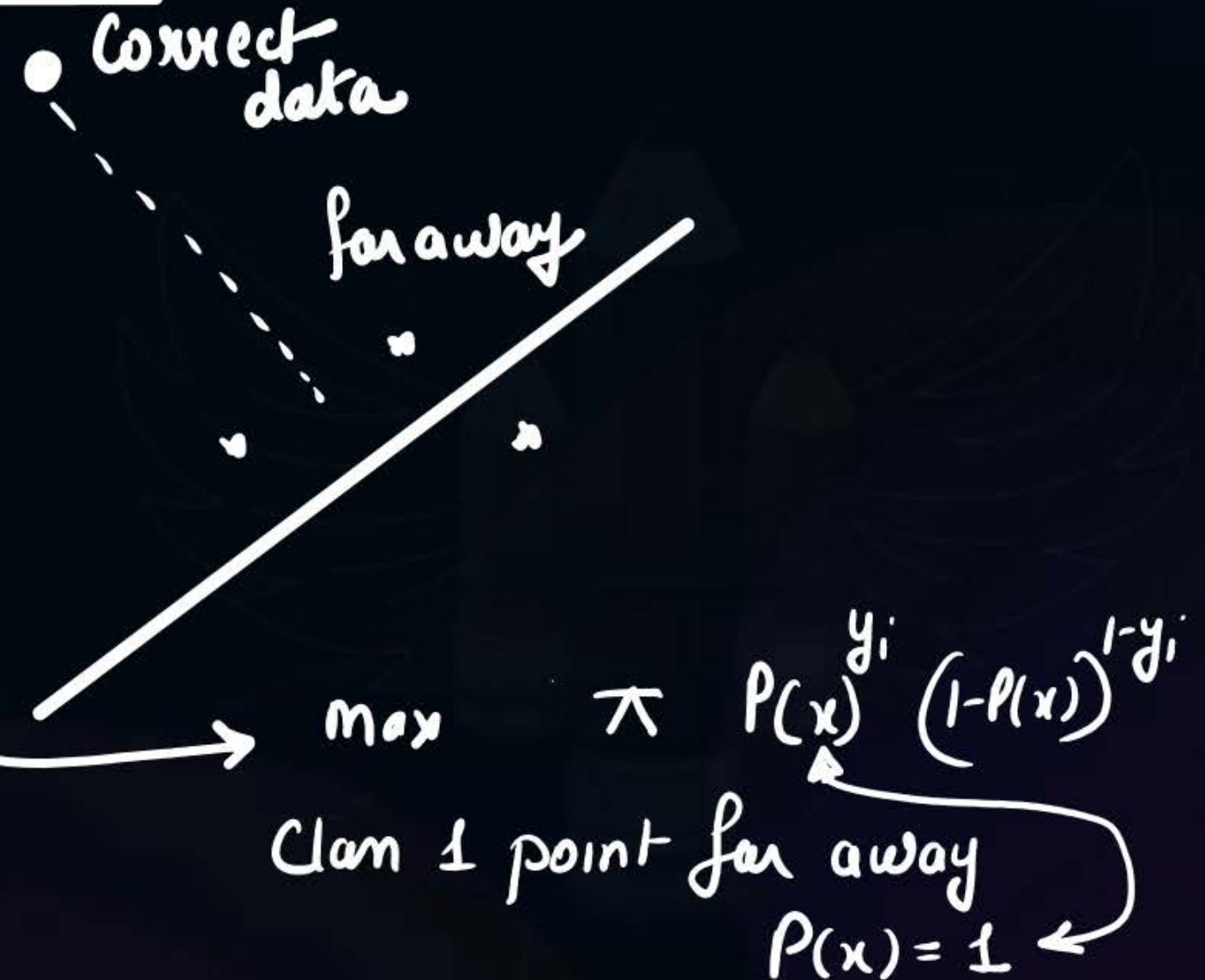
D

✗ For a linearly separable problem, there exists some initialization of the weights which might lead to non-convergent cases.

[MCQ]

#Q. Consider a binary classification problem. Suppose I have trained a model on a linearly separable training set, and now I get a new labeled data point which is correctly classified by the model, and far away from the decision boundary. If I now add this new point to my earlier training set and re-train, in which cases is the learnt decision boundary likely to change?

- A** When my model is a perceptron. ✓
- B** When my model is logistic regression ✓
- C** When my model is an SVM ✓
- D** None of these ✗



$$p(x) = \frac{1}{1 + e^{-(\omega x_i)}}$$

$$\omega x_i \gg 0 \quad p(x) = 1$$

[MCQ]

#Q. The perceptron algorithm will converge:

-  **A** It the data is linearly separable
- B** As long as you initialize θ to all 0's
- C** Even if the data is linearly inseparable
- D** Always

[NAT]

#Q. Suppose we use leave-one-out cross validation meaning we use 7-fold cross validation with a split of 6 to 1 between training set and validating set. Compute the average classification error over the 7- folds.



#Q. Leave-one-out cross-validation (LOQCV) is a special case of k-fold cross validation where:

- ☒ **A** The training set contains all but one sample, and the remaining samples is used for testing.
- ☐ **B** The training set contains only one sample, and the remaining sample is used for testing
- ☐ **C** The training set contains exactly one sample from each class, and the remaining samples are used for testing.
- ☐ **D** The training set contains one sample from each fold, and the remaining folds are used for testing.

[MCQ]

#Q. Why is it important to use a different test set to evaluate the final performance of the model, rather than the validation set used during model selection?

- ☒ **A** The model may have overfit to fit the validation set
- ☒ **B** The test set is a better representation of new, unseen data
- ☒ **C** Both a and b 
- ☐ **D** None of the above

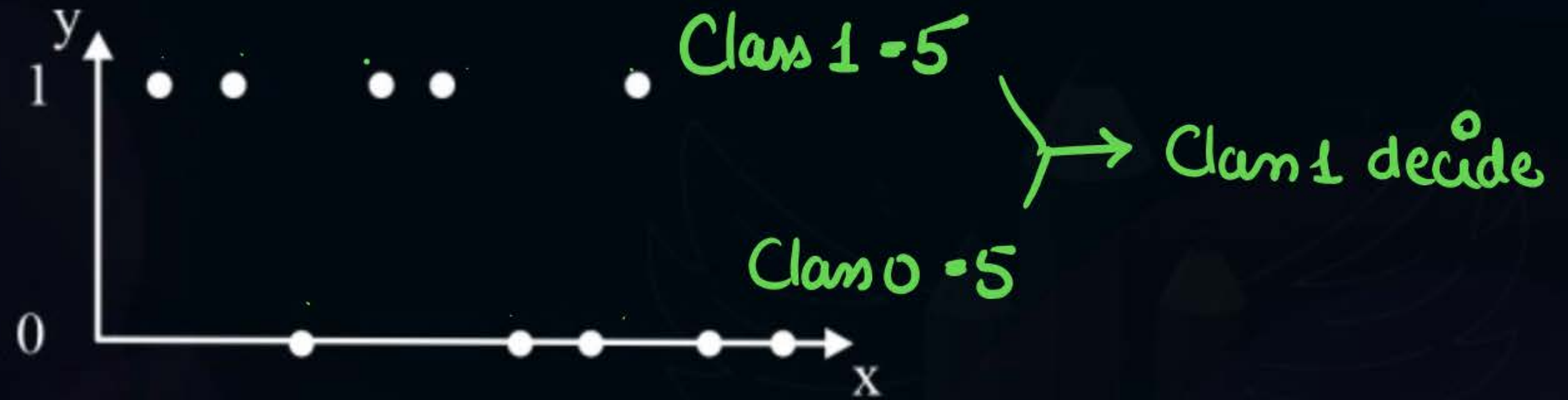
[MCQ]

#Q. Under which of the following conditions is k-fold cross-validation the same as leave-one-out cross-validation?

- A** The training set and test set have the same number of examples
- B** The training set and tuning set have the same number of examples
- C** $k = 1$
- D** $k = n$, where n is the total number of examples
- E** None of the above

[MCQ]

#Q. Suppose you are using a Majority Classifier on the following training set containing 10 examples where each example has one real-valued feature, x , and a binary class label, y , with value 0 or 1. Define this Majority Classifier to predict the class label that is in the majority in the training set, regardless of the input value. In case of ties, predict class 1.



What is the training set accuracy?

A

0%

B

10%

C

50%

D

90%

E

100%

[NAT]

#Q. Consider the following dataset with one real-value input x and one binary output y . We are going to use k -NN with unweighted Euclidean distance to predicted y for x .



X	Y	
-0.1	-	✓ +
0.7	+	+
1.0	+	+
1.6	-	✓ +
2.0	+	+
2.5	+	+
3.2	-	-
3.5	-	-
4.1	+	✓ -
4.9	+	+



3/10 200C ✓

What is the leave-one-out cross validation error of 1-NN on this dataset?
Give your answer as the number of misclassifications.

[MCQ]

#Q. To prevent overfitting of our linear model, we would apply regularization on the model. Which of the following dataset should we evaluate using our model in order to decided the right amount of regularization?

- A** Train
- B** Validation
- C** Test
- D** All of the above

[MCQ]

#Q. What is cross-validation not used for?

- ☒ **A** To evaluate the performance of a machine learning model on unseen data.
- ☒ **B** To select a model's hyperparameters
- ☒ **C** To determine the generalization of a machine learning model.
- ☐ **D** To train multiple machine models on different datasets.

[NAT]

#Q. Consider a single perceptron with sign[↙] activation function. The perceptron is represented by weight vector $[0.4 \ -0.3 \ 0.1]^t$ and a bias $\theta = 0$. If the input vector to the perceptron is $X = [0.2 \ 0.6 \ 0.5]$ then the output of the perceptron is :

$$\begin{aligned}\text{input : } & (0.4 \times 0.2 + 0.6 \times -0.3 + 0.5 \times 0.1) \\ & 0.08 - 0.18 + 0.05 \Rightarrow -0.05 \\ & \text{O/P} \Rightarrow -1 \Leftarrow\end{aligned}$$

#Q. What should be the number of neurons in its input layer, if patterns consisting of 10 attributes each are to be classified by this MLP?

10 ✓

#Q. Consider a Multi-Layer Perceptron (MLP) model with one hidden layer and one output layer. The hidden layer has 12 neurons, and the output layer has 5 neurons. The input to the MLP is a 6-dimensional vector. Each neuron is fully connected to the previous layer, and a bias term is included for each neuron. The activation function used is ReLU for the hidden layer and Softmax for the output layer.

Calculate the total number of trainable parameters in this MLP model.

$$\begin{array}{ccc} 6 \times 12 & 12 \times 5 & +12+5 \\ 6 & 12 & 5 \\ \text{Ip} & \text{HD} & \text{Op} \end{array} \quad \text{149}$$

[NAT]

- #Q. Consider an MLP model with three hidden layers and one output layer. The first hidden layer has 20 neurons, the second has 12 neurons, the third has 8 neurons, and the output layer has 3 neurons. The input to the MLP is a 10-dimensional vector. Each neuron is connected to all neurons in the previous layer, and each neuron includes a bias term. Determine the total number of trainable parameters in this MLP model.

10 20 20 12 12 8 8 3 3

200 240 96 24

603

[NAT]

#Q. Consider a single-layer perceptron with a sign activation function. The perceptron is represented by the weight vector $[0.5, -0.2, 0.3]^T$ and bias $\theta =$ ~~1~~
the input vector to the perceptron is: .5

$$X = [0.4, 0.7, 0.2]$$

Then, determine the output of the perceptron.

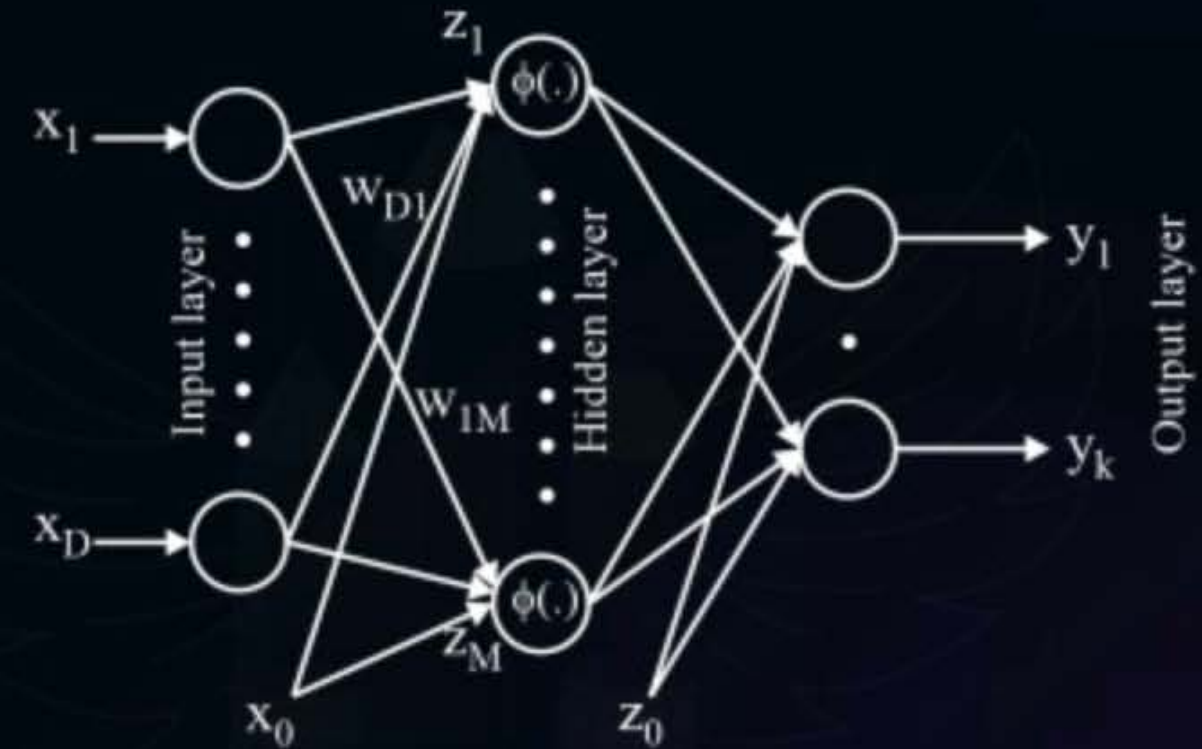
$$I/p = (.2 - .14 + .06 + .5) = .5$$

$$I/p > 0, O/p \Rightarrow +1 \checkmark$$

[MCQ]

#Q. Consider a two-layer network of the form shown in Figure 1 with M hidden units having $\phi(.) = \tanh(.)$ activation functions and full connectivity in both layers. If we change the sign of all of the weights and the bias feeding into a particular hidden unit, then, for a given input pattern

Network diagram for the two layer neural network corresponding to questions 1, 2, 3 and 4. The input, hidden, and output variables are represented by nodes, and the weight parameters are represented by links between the nodes, in which the bias parameters are denoted by links coming from additional input and hidden variables x_0 and z_0 . Arrows denote the direction of information flow through the network during forward propagation.

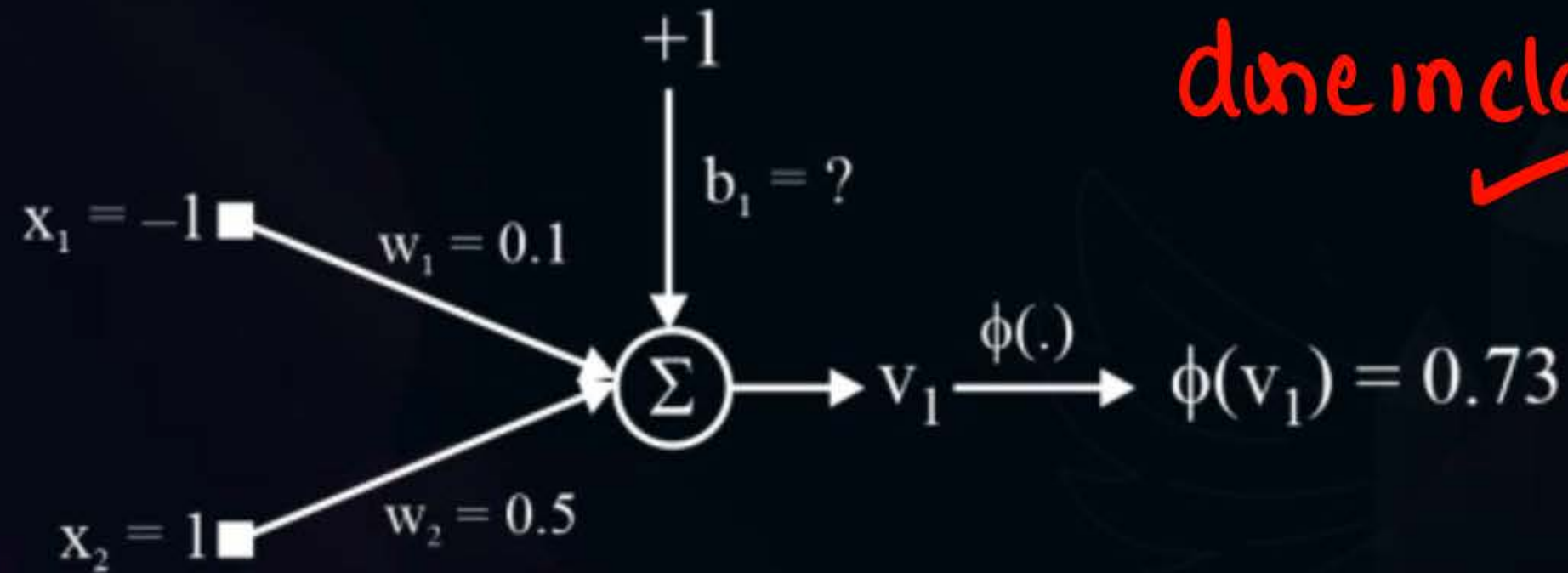


- A** The sign of the activation of the hidden unit will be reversed.
- B** The sign of the activation of the hidden unit will remain same.
- C** The output of the hidden unit will be zero irrespective of the input.
- D** The output of the hidden unit will be scaled by a positive integer.

≡
done in
clam

[NAT]

- #Q. Consider a two input neuron with logistic activation function with slope parameter $a = 2$. Let the inputs be $[-1, 1]$ and the weights are $[0.1, 0.5]$ respectively. The output of the neuron is 0.73 . The value of the bias b_1 is (Round bias value till 1 decimal point)?



done in class ✓

[MCQ]

#Q. Recall that the output of a perceptron is 0 or 1. For each of the two following data sets, select the perceptron network with the fewest nodes that will separate the classes. You can use the same network more than once

A (i)- A; (ii)-B

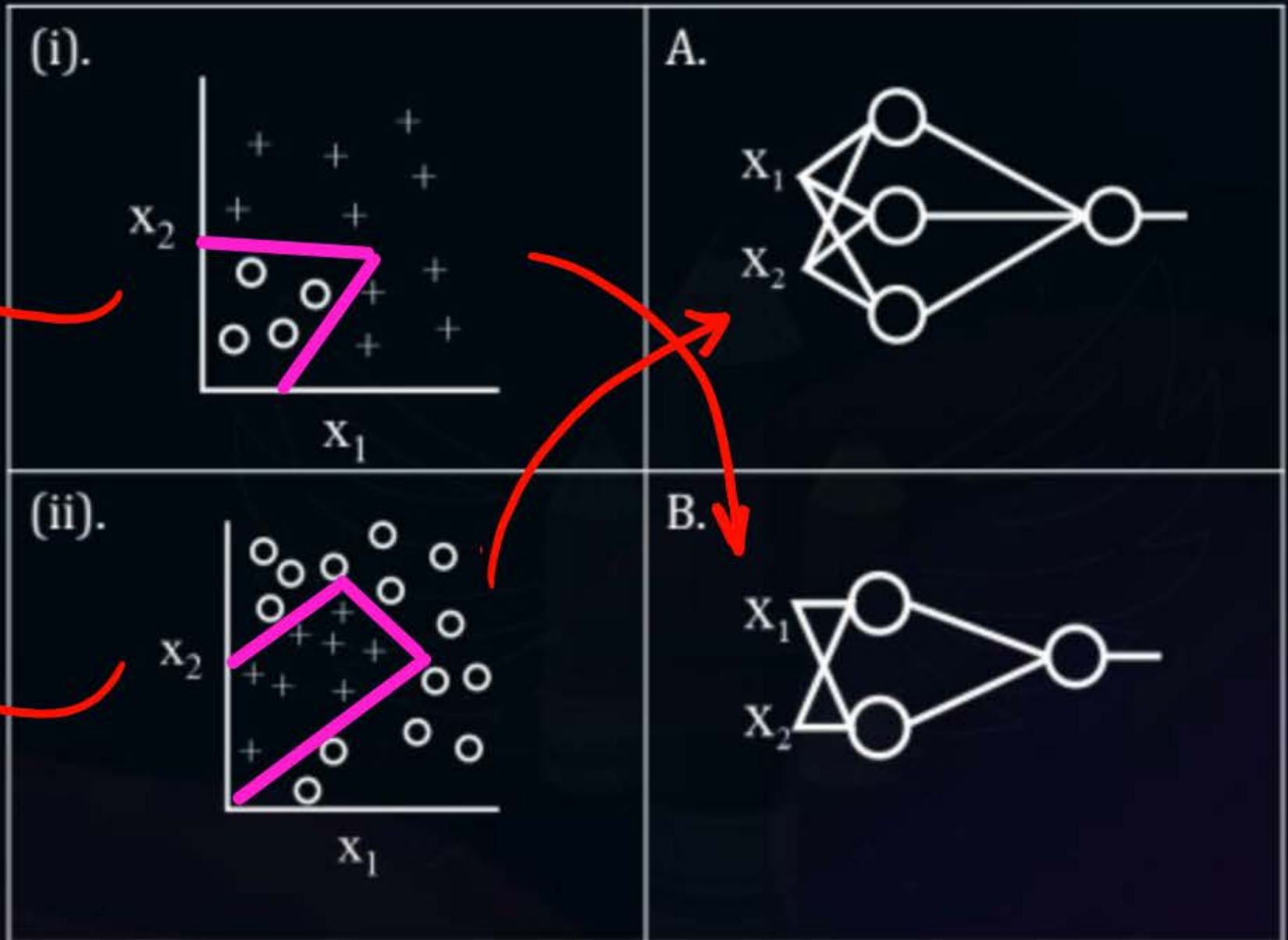
B (i)- B; (ii)-A

C (i)- A; (ii)-A

D (i)- B; (ii)-B

Clarify
2 Boundary

3 boundary



[MCQ]

#Q. Let's say you have a MLP network which is trained on 5-dimensional input data. The network has 3 hidden layers, with 50, 30, and 20 nodes respectively. The network is trained to predict a 10-way classification task using softmax activation. If we ignore bias terms, how many weights does this neural network have?

A

2,550

B

1,500,000

C

115

D

30,000

250 1500 600 2000
5 50 30 20 10 output

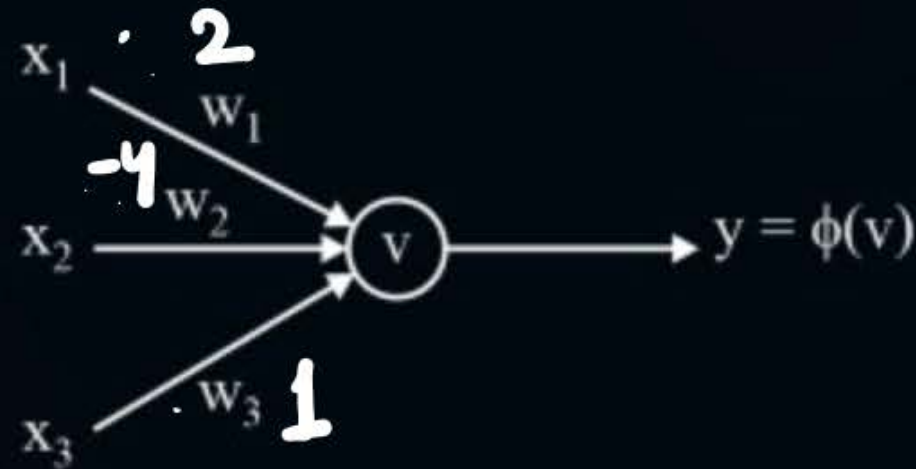
$$250 + 1500 + 600 + 200 = 2550$$

#Q. Consider the unit shown on Figure 1. Suppose that the weights corresponding to the three inputs have the following values

$$w_1 = 2$$

$$w_2 = -4$$

$$w_3 = 1$$



and the activation of the unit is given by the step-function:

$$\phi = \begin{cases} 1 & v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Calculate what will be the sum of output value y of the unit for each of the following input patterns:

Pattern	P ₁	P ₂	P ₃	P ₄
x ₁	1	0	1	1
x ₂	0	1	0	1
x ₃	0	1	1	1

I/P₁
↓
2
↓
1

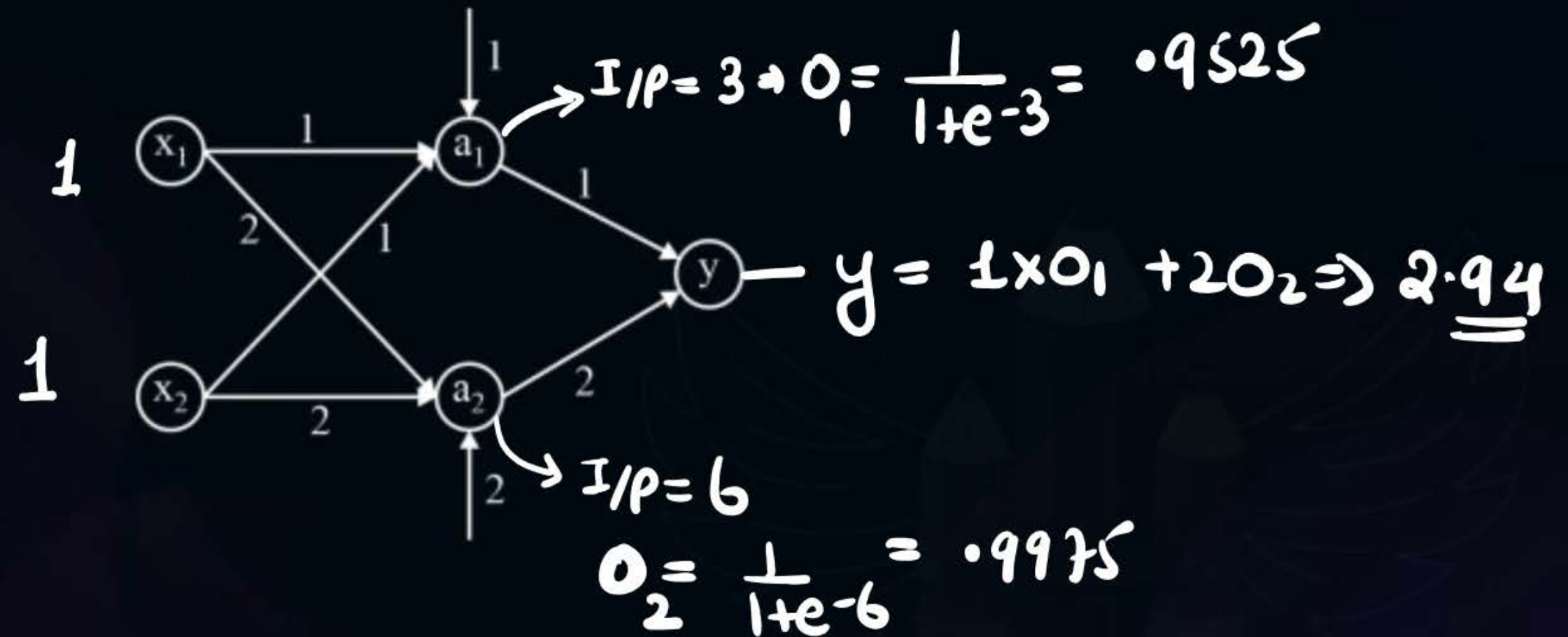
I/P₂
↓
-3
↓
0

I/P₃
↓
3
↓
1

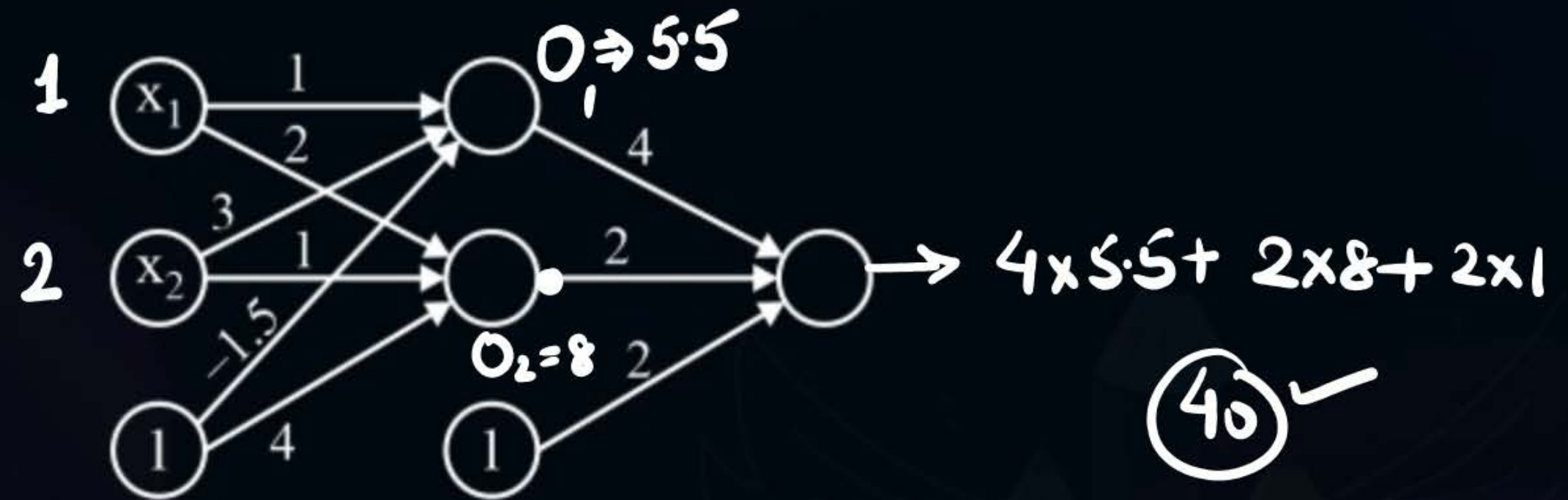
I/P₄
↓
-1
↓
0

[NAT]

- #Q. What is the output(round to 2 decimal points) of the network for $x_1 = 1$, $x_2 = 1$? Assume sigmoid is being at a_1 , a_2 and at y no activation function is used.



#Q. What is the output of the network for $x_1 = 1$, $x_2 = 2$. Assume at every level identity function is used.



- #Q. Consider an MLP model with two hidden layers and one output layer. The first hidden layer has 8 neurons, the second hidden layer has 5 neurons, and the output layer has 3 neurons. The input to the MLP is a 4-dimensional vector. Each neuron is fully connected to every neuron in the previous layer, and a bias term is included for each neuron. Calculate the total number of trainable parameters in this MLP model.

$$\begin{array}{cccc}
 & 32 & 40 & 15 \\
 4 & 8 & 5 & 3 \\
 & 8 & 5 & 3
 \end{array}
 \Rightarrow 103$$

[MCQ]

#Q. What is the function of the hidden layers in a multilayer perceptron (MLP)?

- ☒ **A** Hidden layers perform feature extraction and transformation
- ☐ **B** Hidden layers provide direct access to the input data
- ☐ **C** Hidden layers perform the final classification or regression task
- ☐ **D** Hidden layers are not necessary in an MLP

[MCQ]

#Q. Consider the following statements:

S_1 : The Bias-Variance Trade-Off helps in balancing the complexity of the model and its performance on unseen data. ✓

S_2 : A Multi-Layer Perceptron (MLP) can only approximate linear decision boundaries. ✗

Which of the above is true?

A Only S_1 ✓

B Only S_2

C Both S_1 and S_2

D None of these



THANK - YOU